

Reasoning: Direction Sense Test - Complete Study Material

Theory, 12 Solved Examples, and 25 Practice Questions with Answer Key

Part 1: Theory (Important Rules of Direction Sense)

1. Four Main Directions

There are four main directions: **North (N)**, **South (S)**, **East (E)**, and **West (W)**.

2. Four Cardinal Directions (Sub-directions)

- **North-East (NE)**: Between North and East
- **North-West (NW)**: Between North and West
- **South-East (SE)**: Between South and East
- **South-West (SW)**: Between South and West

3. Turns and Angles

- **Right Turn**: Equivalent to a 90° clockwise turn.
- **Left Turn**: Equivalent to a 90° anti-clockwise (counter-clockwise) turn.
- **Opposite Directions**: North faces South, East faces West.

4. Pythagoras Theorem (Shortest Distance)

To find the shortest distance between the starting point and the final destination, we use the right-angled triangle property:

$$(Hypotenuse)^2 = (Base)^2 + (Perpendicular)^2$$

$$Distance = \sqrt{(Base)^2 + (Perpendicular)^2}$$

Part 2: Solved Examples (12 Examples)

Example 1: A man walks 5 km towards North, then turns right and walks 3 km. Then he turns right again and walks 5 km. Which direction is he facing now?

Solution: He starts walking North. First right turn makes him face East. Second right turn makes him face South.

Answer: South

Example 2: Rohan walks 10 m North, turns left and walks 10 m, turns left again and walks 10 m. How far and in which direction is he from his starting point?

Solution: Moving 10 m North and then 10 m West, and finally 10 m South brings him directly West of the starting point at a distance of 10 m.

Answer: 10 m West

Example 3: A cat runs 20 m to the East, turns right and runs 10 m, turns right again and runs 20 m. Finally, it turns left and runs 5 m. Which direction is it running now?

Solution: Facing East -> Right turn (South) -> Right turn (West) -> Left turn (South).

Answer: South

Example 4: From a point, Sita walks 3 km North, turns West and walks 4 km. What is the shortest distance between her starting point and final position?

Solution: Using Pythagoras theorem: $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ km.

Answer: 5 km

Example 5: If South-East becomes North, North-West becomes South, and all other directions change in the same manner, what will West become?

Solution: Directions rotate by 135° clockwise or counter-clockwise. West becomes South-East.

Answer: South-East

Example 6: One morning, Rahul and Sachin were talking facing each other. Sachin's shadow fell exactly to the right of Rahul. Which direction was Rahul facing?

Solution: In the morning, the sun is in the East, so shadows fall towards the West. Since Sachin's shadow fell to Rahul's right, Rahul must be facing North.

Answer: North

Example 7: A boy goes 15 meters towards West, turns North and goes 20 meters, turns East and goes 15 meters. How far is he from his starting point?

Solution: The West movement of 15m cancels out the East movement of 15m. Only the North movement of 20m remains.

Answer: 20 meters

Example 8: Starting from point P, Amit walked 10 m East, turned South and walked 10 m, turned West and walked 10 m. How far is he from point P?

Solution: 10 m East and 10 m West cancel horizontal displacement, leaving 10 m South.

Answer: 10 m South

Example 9: A clock shows 3 o'clock. If the minute hand points towards North-East, which direction does the hour hand point?

Solution: At 3 o'clock, the minute hand is at 12 (North) and hour hand is at 3 (East). If North becomes North-East, East becomes South-East.

Answer: South-East

Example 10: Kiran walks 6 km towards East, then turns left and walks 8 km. What is the shortest distance from the starting point?

Solution: $\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ km.

Answer: 10 km

Example 11: A person walks 12 km towards North, then 5 km towards West. How far is he from the starting point?

Solution: $\sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$ km.

Answer: 13 km

Example 12: Priya starts from her home, walks 3 km West, turns right and walks 4 km to reach her school. In which direction and at what distance is her school from home?

Solution: Distance = $\sqrt{3^2 + 4^2} = 5$ km. Direction is North-West with respect to home.

Answer: 5 km North-West

Part 3: Practice Questions (25 Questions)

Q1. If you are facing North and turn 90° clockwise, which direction are you facing?

- (A) South (B) East (C) West (D) North-East

Q2. A man walks 5 km East, then turns South and walks 12 km. What is his distance from the start?

- (A) 13 km (B) 17 km (C) 7 km (D) 15 km

Q3. Which of the following is a cardinal direction?

- (A) North-East (B) South-West (C) North (D) North-West

Q4. If North becomes South-East, what does West become?

- (A) North-East (B) South-West (C) North-West (D) South-East

Q5. Rahul walks 10 m North, turns right and walks 10 m. Which direction is he facing?

- (A) North (B) South (C) East (D) West

Q6. A dog runs 4 km North, then 3 km East. How far is it from the starting point?

- (A) 7 km (B) 5 km (C) 6 km (D) 4 km

Q7. If you walk 2 km West, then turn left and walk 2 km, which direction are you heading?

- (A) South (B) North (C) East (D) West

Q8. A person facing South turns 135° anti-clockwise. Which direction is he facing now?

- (A) North-West (B) North-East (C) South-East (D) South-West

Q9. From point A, Tina walks 8 m South, turns West and walks 6 m. Shortest distance?

- (A) 14 m (B) 10 m (C) 12 m (D) 8 m

Q10. If East is called North, North is called West, West is called South, what is South called?

- (A) North (B) West (C) East (D) South

Q11. A cyclist goes 30 km North, turns East and goes 40 km. Total displacement from start?

- (A) 70 km (B) 50 km (C) 10 km (D) 60 km

Q12. Which direction is opposite to North-East?

- (A) South-East (B) North-West (C) South-West (D) South

Q13. Aman walks 5 m straight, turns right and walks 3 m, turns right again and walks 5 m. Direction?

- (A) Opposite to start (B) Same as start (C) Perpendicular (D) None of these

Q14. A clock shows 9 o'clock. If minute hand points West, hour hand points:

- (A) North (B) South (C) East (D) West

Q15. Two persons start from the same point. A walks 3 km North, B walks 4 km East. Distance between them?

- (A) 7 km (B) 5 km (C) 1 km (D) 12 km

Q16. If South-West becomes East, North-East becomes:

- (A) West (B) North (C) South (D) East

Q17. Rohit walks 20 m towards South, turns left and walks 15 m. Direction from start?

- (A) South-West (B) South-East (C) North-East (D) North-West

Q18. A building faces East. If you walk out of the front door and turn left, which direction do you go?

- (A) South (B) North (C) West (D) East

Q19. A person walks 9 km North, 12 km East, then 9 km South. Distance from start?

- (A) 30 km (B) 12 km (C) 21 km (D) 15 km

Q20. What angle is formed between North and South-East directions?

- (A) 90° (B) 135° (C) 180° (D) 45°

Q21. Mohan walks 5 km South, turns left and walks 5 km. Direction from start?

- (A) South-East (B) South-West (C) North-East (D) North-West

Q22. If North is West, West is South, South is East, what is East?

- (A) South (B) North (C) West (D) East

Q23. A car travels 10 km North, turns left and travels 10 km, turns left again and travels 10 km. Position?

- (A) 10 km West of start (B) At start (C) 10 km East of start (D) 20 km North

Q24. How many degrees does the minute hand move in 15 minutes?

- (A) 45° (B) 90° (C) 180° (D) 360°

Q25. Sonu walks 1 km East, then 2 km South, then 1 km West. Distance from start?

- (A) 4 km (B) 2 km (C) 1 km (D) 3 km

Answer Key for Practice Questions

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
Q1	(B) East	Q8	(A) North-West	Q15	(B) 5 km	Q22	(B) North
Q2	(A) 13 km	Q9	(B) 10 m	Q16	(A) West	Q23	(A) 10 km West
Q3	(C) North	Q10	(C) East	Q17	(B) South-East	Q24	(B) 90°
Q4	(A) North-East	Q11	(B) 50 km	Q18	(B) North	Q25	(B) 2 km
Q5	(C) East	Q12	(C) South-West	Q19	(D) 15 km	-	-
Q6	(B) 5 km	Q13	(A) Opposite to start	Q20	(B) 135°	-	-
Q7	(A) South	Q14	(A) North	Q21	(A) South-East	-	-